

Triangles, radians and sectors - Set 2

TMUA practice questions by topic. Specification MM4.1, MM4.2, M5.16. 5 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 A circle has radius r . A sector of angle θ radians has the same area as the segment cut off by a chord that subtends an angle of $\frac{\pi}{2}$ at the centre. Find θ .
- A $\frac{\pi}{2} - 1$
B $\frac{\pi}{4} - \frac{1}{2}$
C $\frac{\pi}{4} - \frac{1}{2}$
D 1
E $\pi - 2$
F $\frac{\pi}{2} - \frac{1}{2}$
- 2 A triangle has sides a, b, c with $a^2 + b^2 < c^2$. Which of the following statements is/are true?
- I The angle opposite c is obtuse.
II c is the longest side.
III The triangle cannot be isosceles.
- A none of them
B I only
C II only
D III only
E I and II only
F I and III only
G II and III only
H I, II and III
- 3 A vertical pole BT stands on horizontal ground. From a point A on the ground the angle of elevation of T is 30° ; from a point C on the ground, 20 m closer to the pole than A and in line with A and B , the angle of elevation is 60° . Find the height of the pole.
- A 10 m
B $20\sqrt{3}$ m
C 15 m
D $10\sqrt{3}$ m
E $\frac{20}{\sqrt{3}}$ m
F 30 m

- 4 Triangle ABC has $AB = 7$, $BC = 8$, $CA = 9$. Find the area of the triangle.
- A 28
B $\sqrt{720}$
C $24\sqrt{5}$
D $12\sqrt{5}$
E 30
F $6\sqrt{5}$
- 5 Two circles of radius 2 have their centres a distance 2 apart. Find the area of the region common to both circles.
- A $\frac{2\pi}{3}$
B $\frac{4\pi}{3} - 2\sqrt{3}$
C $\frac{8\pi}{3} - 2\sqrt{3}$
D $\frac{8\pi}{3} - 4\sqrt{3}$
E $2\pi - 2\sqrt{3}$
F $\frac{4\pi}{3} - \sqrt{3}$